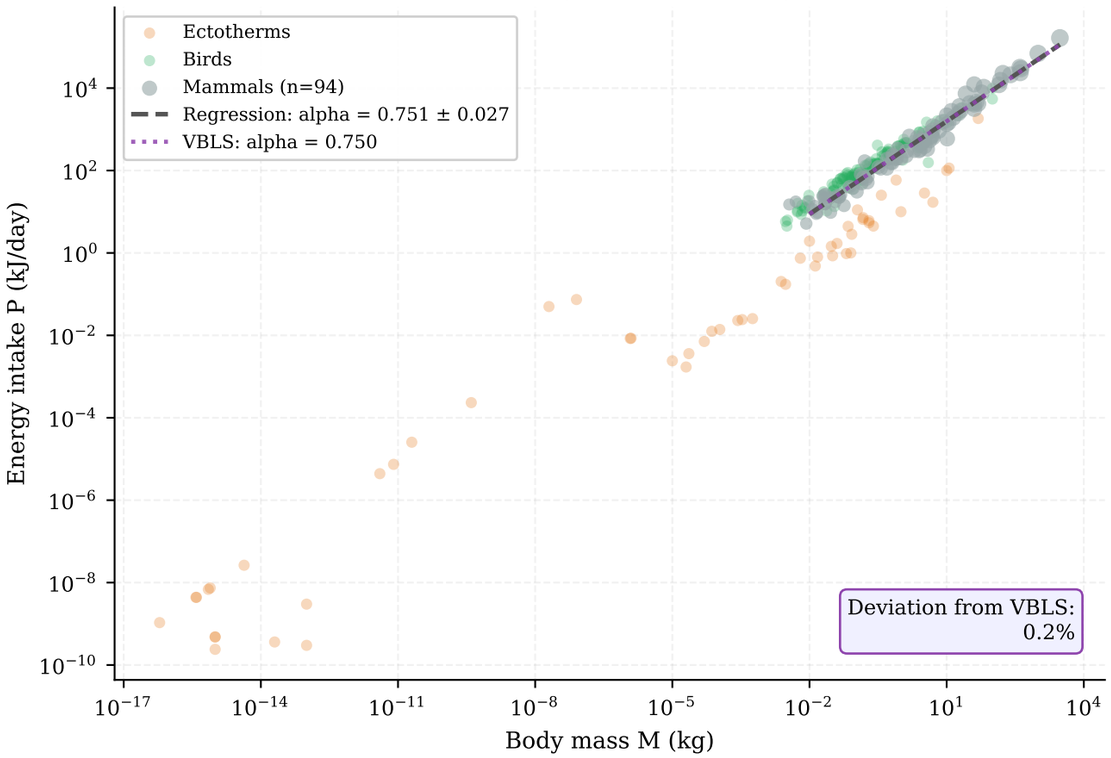
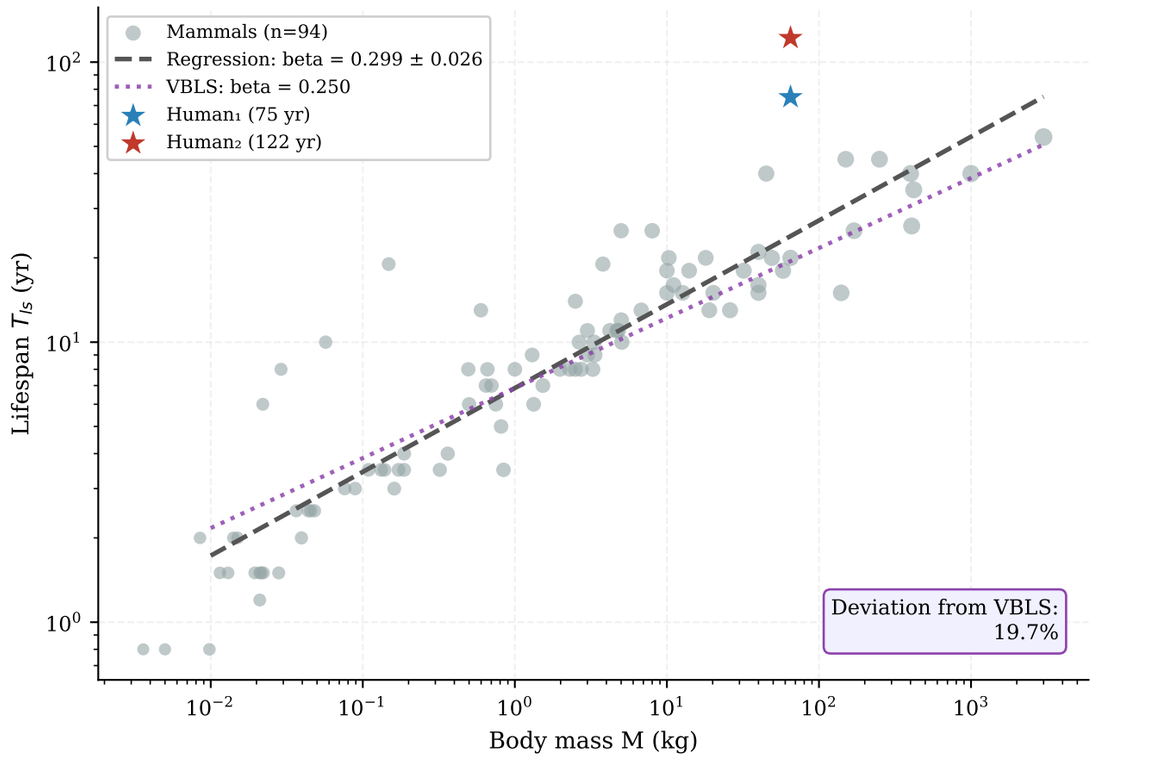
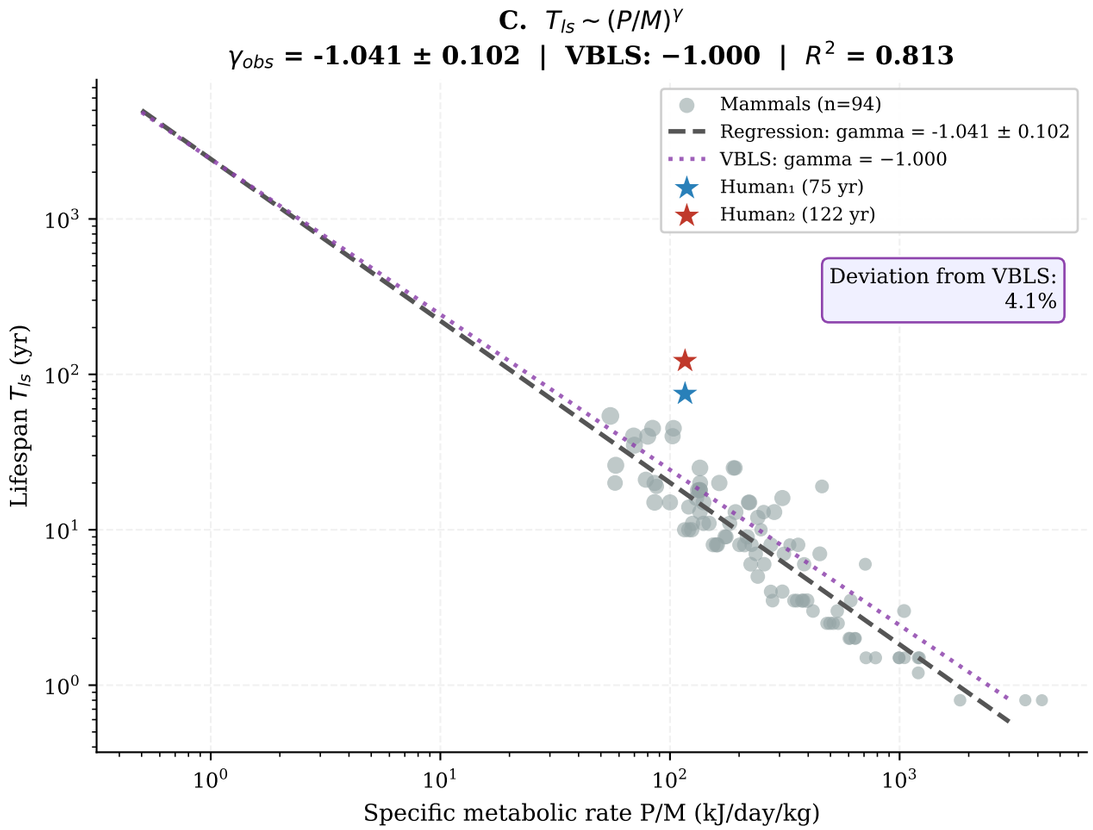
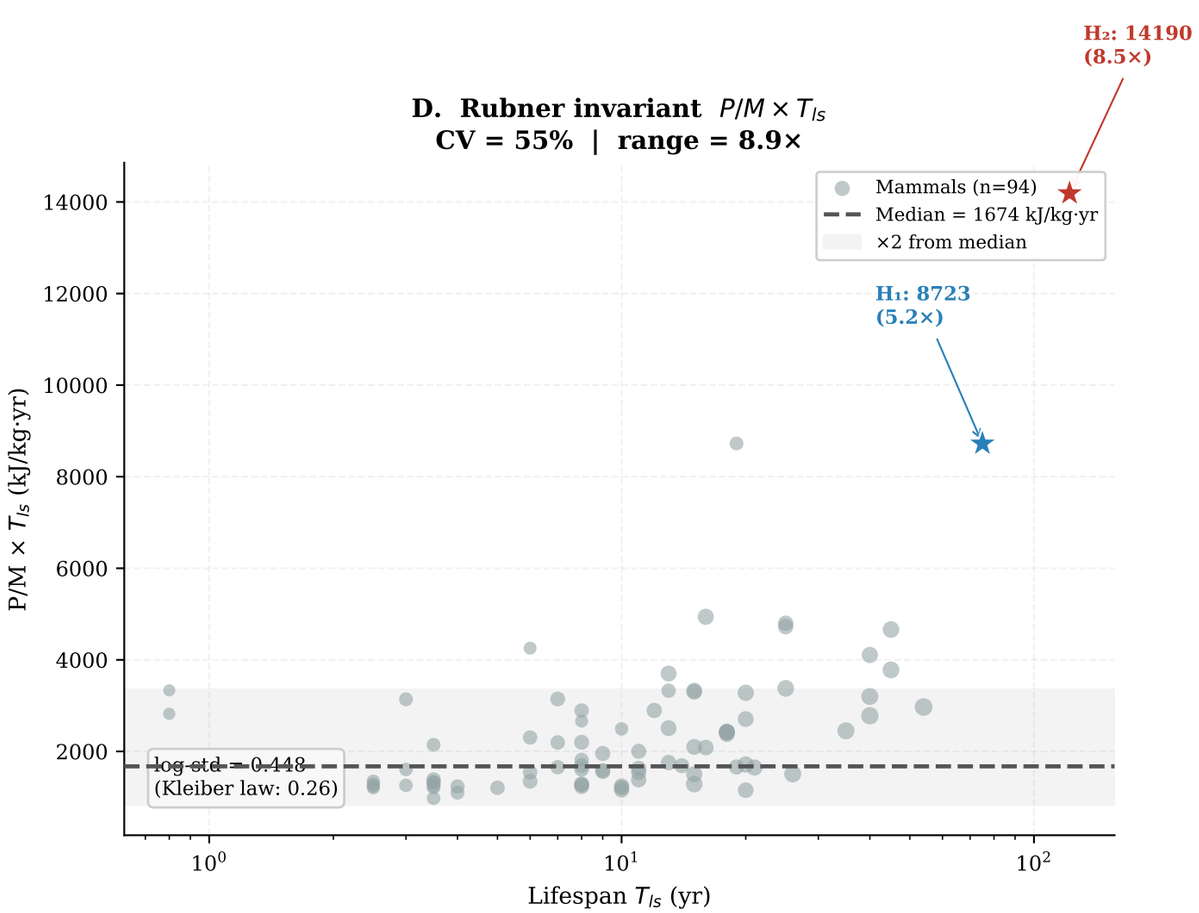

A. Kleiber's law $P \sim M^{\alpha}$
α_{obs} = 0.751 ± 0.027 | VBLS: 0.750 | R^2 = 0.970
Ectotherms
Birds
Mammals (n=94)
Regression: alpha = 0.751 ± 0.027
VBLS: alpha = 0.750
Deviation from VBLS: 0.2%
Energy intake P (kJ/day)
Body mass M (kg)


B. Longevity allometry $T_{ls} \sim M^{\beta}$
β_{obs} = 0.299 ± 0.026 | VBLS: 0.250 | R^2 = 0.849
Mammals (n=94)
Regression: beta = 0.299 ± 0.026
VBLS: beta = 0.250
Human₁ (75 yr)
Human₂ (122 yr)
Deviation from VBLS: 19.7%
Lifespan T_{ls} (yr)
Body mass M (kg)


C. $T_{ls} \sim (P/M)^{\gamma}$
γ_{obs} = -1.041 ± 0.102 | VBLS: −1.000 | R^2 = 0.813
Mammals (n=94)
Regression: gamma = -1.041 ± 0.102
VBLS: gamma = −1.000
Human₁ (75 yr)
Human₂ (122 yr)
Deviation from VBLS: 4.1%
Lifespan T_{ls} (yr)
Specific metabolic rate P/M (kJ/day/kg)


D. Rubner invariant $P/M \times T_{ls}$
CV = 55% | range = 8.9×
H₂: 14190 (8.5×)
H₁: 8723 (5.2×)
Mammals (n=94)
Median = 1674 kJ/kg·yr
×2 from median
log std = 0.448
(Kleiber law: 0.26)
P/M × T_{ls} (kJ/kg·yr)
Lifespan T_{ls} (yr)